CipherShare: Advancing Cloud Storage Security SLAs Through Browser-Native Hybrid Cryptography and Non-Repudiable Attestations

An Applied Implementation and Modern Extension of Popa et al. (CloudProof, USENIX ATC '11)
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Abstract

Cloud storage providers such as Amazon Web Services (AWS S3) and Microsoft Azure offer Service Level Agreements (SLAs) guaranteeing high availability (e.g., 99.9%), yet they universally fail to provide cryptographically enforceable **security guarantees**. In their landmark paper *CloudProof* (USENIX ATC 2011), Raluca Ada Popa et al. established that clients must not only detect cloud security misbehavior (violations of Confidentiality, Integrity, Write-Serializability, and Freshness), but also possess the capability to cryptographically **prove these violations to an objective third party** without enabling clients to frame an honest cloud.

This report presents **CipherShare**, a modern software architecture that implements, evaluates, and significantly improves upon the foundation laid by Popa et al. While CloudProof demonstrated the viability of security SLAs, its 2011 implementation suffered from critical contemporary vulnerabilities and architectural friction: reliance on deprecated primitives (**SHA-1** and **1024-bit RSA**), absence of authenticated associated data (**AAD**) binding, reliance on a native desktop **C# / .NET** binary, and lack of browser-isolated private key storage.

CipherShare modernizes CloudProof into a zero-installation, browser-native cloud storage architecture. We replace unauthenticated AES-CTR with **AES-256-GCM** incorporating **Canonical AAD contextual binding**, upgrade key encapsulation to **RSA-OAEP 3072-bit**, protect client private keys in browser **IndexedDB** using **PBKDF2-SHA-256 (600,000 iterations)** + **AES-GCM**, and implement real-time **non-repudiable attestation exchanges** over modern cloud object storage (Supabase S3). Empirical benchmarks show our browser-based engine achieves **216 MB/s** symmetric throughput with negligible latency overhead (**4.62 ms** on 1 MB payloads), successfully detecting single-bit corruptions and enforcing strict **HTTP 403 Forbidden** isolation against unauthorized access.

1. Problem Statement & The Cloud Security SLA Dilemma

1.1 The Inadequacy of Modern Cloud SLAs

Contemporary enterprises entrust petabytes of confidential records, source repositories, and proprietary financial files to public commodity clouds. However, commercial cloud SLAs remain purely availability-centric:

- **AWS S3 SLA**: Reimburses contractual credits only if availability dips below 99.9%.

- **Microsoft Azure SLA**: Covers downtime, but provides **zero contractual guarantees** regarding data confidentiality, bit-level integrity, or access control violations.

These vulnerabilities are documented by historical cloud failures:

1. **Silent Bit Corruption**: In June 2008, Amazon S3 suffered internal load balancer corruptions that silently altered single bytes in stored files, causing customer hash verifications to fail intermittently under load.
2. **Access Control Leaks**: Google Docs suffered access control flaws resulting in private documents being shared with unintended third parties.
3. **Catastrophic Administrative Loss**: Cloud provider LinkUp (MediaMax) lost 45% of customer data permanently due to administrator error, forcing the firm out of business.

1.2 The "Encryption at Rest" Fallacy

Cloud vendors market "Server-Side Encryption with KMS (SSE-KMS)" as an enterprise security panacea. However, in server-side encryption: $$\text{Plaintext } P \xrightarrow{\text{TLS}} \text{Cloud Server RAM } [P, K_{\text{KMS}}] \xrightarrow{\text{Encrypt}} \text{Storage Disk } [C]$$ The cloud server retains full access to content-encryption keys and unencrypted buffers in volatile memory. A hypervisor breach, malicious cloud employee, rogue subprocess, or government subpoena yields total plaintext access.

1.3 Popa et al.'s Four Pillars of Cloud Storage Security

Popa et al. formalized the four mandatory security properties that an accountable cloud storage service must uphold:

1. **Confidentiality (C)**: Illegitimate users and the cloud provider itself must be mathematically incapable of reading block contents.
2. **Integrity (I)**: Every read must return the exact content written by an authorized user; unauthorized bit modifications or server substitutions must be detected.
3. **Write-Serializability (W)**: Updates must follow a total, unambiguous order, preventing fork attacks where the cloud serves divergent views to different clients.
4. **Read Freshness (F)**: Reads must return the data committed by the most recent write, preventing stale data replay.

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2. Theoretical Grounding: Analysis of Popa et al. (CloudProof)

Popa et al. proposed **CloudProof** to bridge the gap between untrusted cloud platforms and financial SLAs through an **Attestation Exchange Protocol**.

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Popa et al. Chained Attestation Model (Eq. 1):
chain_hash_i = Hash(data_i || chain_hash_{i-1})
```

2.1 The Attestation Exchange Mechanism

In CloudProof, every cloud interaction generates a cryptographically signed receipt:

- **Client Put Attestation**: Emitted by the writer upon submitting an update, certifying: "*I request committing version v with digest $H(P)$*".

- **Cloud Put Attestation**: Emitted by the cloud acknowledging receipt: "*I certify committing version v into the storage tree*".
- **Cloud Get Attestation**: Emitted by the cloud upon retrieval: "*I certify that this block represents the fresh state at version v for request nonce N*".

2.2 Proofs of Security Violations

CloudProof was the first system enabling clients to construct verifiable **Proofs of Violation**:

- **Proof of Integrity (I) Violation**: A client presents a block where the cloud's signature covers a hash that fails to match the block's actual content.
- **Proof of Write-Serializability (W) Violation**: If the cloud attempts a fork attack, the client presents two cloud put attestations acknowledging the same version number $m+1$ with differing content hashes (Theorem 1 in Popa et al.).
- **Proof of Freshness (F) Violation**: If the cloud returns stale data, the broken sequence in the chained hash function proves that a committed write occurred prior to the get attestation (Theorem 2 in Popa et al.).

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3. Critical Gaps in Popa et al. & CipherShare's Improvements

While groundbreaking, Popa et al.'s design was constrained by the cryptographic primitives and cloud tooling of 2011. **CipherShare addresses six fundamental limitations**:

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+-----------------------------------------------------------------------------------------------+
|                              ARCHITECTURAL & CRYPTOGRAPHIC COMPARISON                          |
+---------------------------+--------------------------------------+-----------------------------------+
| Feature Dimension         | Popa et al. (CloudProof 2011)        | CipherShare (Our Modernized Sys.) |
+---------------------------+--------------------------------------+-----------------------------------+
| Hash Standard             | SHA-1 (Broken, collision-prone)      | SHA-256 / SHA-512 (FIPS 180-4)    |
| Asymmetric Key Size       | 1024-bit RSA (NIST Deprecated)       | RSA-OAEP 3072-bit (128-bit Sec.)  |
| Symmetric Cipher Mode     | AES in Counter Mode (AES-CTR)        | AES-256-GCM (NIST SP 800-38D)     |
| Integrity Verification    | Separate RSA signature per block     | Integrated 128-bit GMAC Tag       |
| Context Binding (AAD)     | None (Vulnerable to transplant)      | Canonical JSON AAD Binding        |
| Execution Environment     | Native C# .NET Desktop Application   | Zero-Install W3C Web Crypto API   |
| Endpoint Key Storage      | Unspecified OS Keyring               | PBKDF2-SHA-256 in IndexedDB Vault |
| Key Distribution          | Broadcast Encryption (Block Families)| Granular Zero-Overlap Envelopes   |
| Violation Verification    | Monthly offline batch epochs         | Real-Time Interactive HTTP Proofs |
+---------------------------+--------------------------------------+-----------------------------------+
```

Detailed Improvement Analysis:

1. **Modern Authenticated Encryption (AEAD) vs AES-CTR**: Popa et al. utilized AES-CTR for encryption and a separate RSA signature for integrity. This introduced a 15% write latency penalty. CipherShare implements **AES-256-GCM**, which computes a 128-bit Galois Message Authentication Code (GMAC) concurrently with encryption in a single pass over the hardware-accelerated polynomial field $\text{GF}(2^{128})$.
2. **Elimination of Ciphertext Transplant Attacks (Canonical AAD)**: In CloudProof, a malicious storage operator could swap encrypted block A into block B's storage slot without detection if both used identical group keys. CipherShare binds the metadata directly into the GMAC tag: $$\text{AAD} = \text{CanonicalJSON}({ \text{fileId}, \text{senderId}, \text{recipientId}, \text{classification}, \text{version} })$$ Altering a single byte of metadata renders the GMAC tag mathematically invalid.
3. **Zero-Installation Browser Execution**: CloudProof required compiling 4,000 lines of C# against the Windows Server 2008 SDK. CipherShare executes directly inside the end-user's browser using the standardized W3C

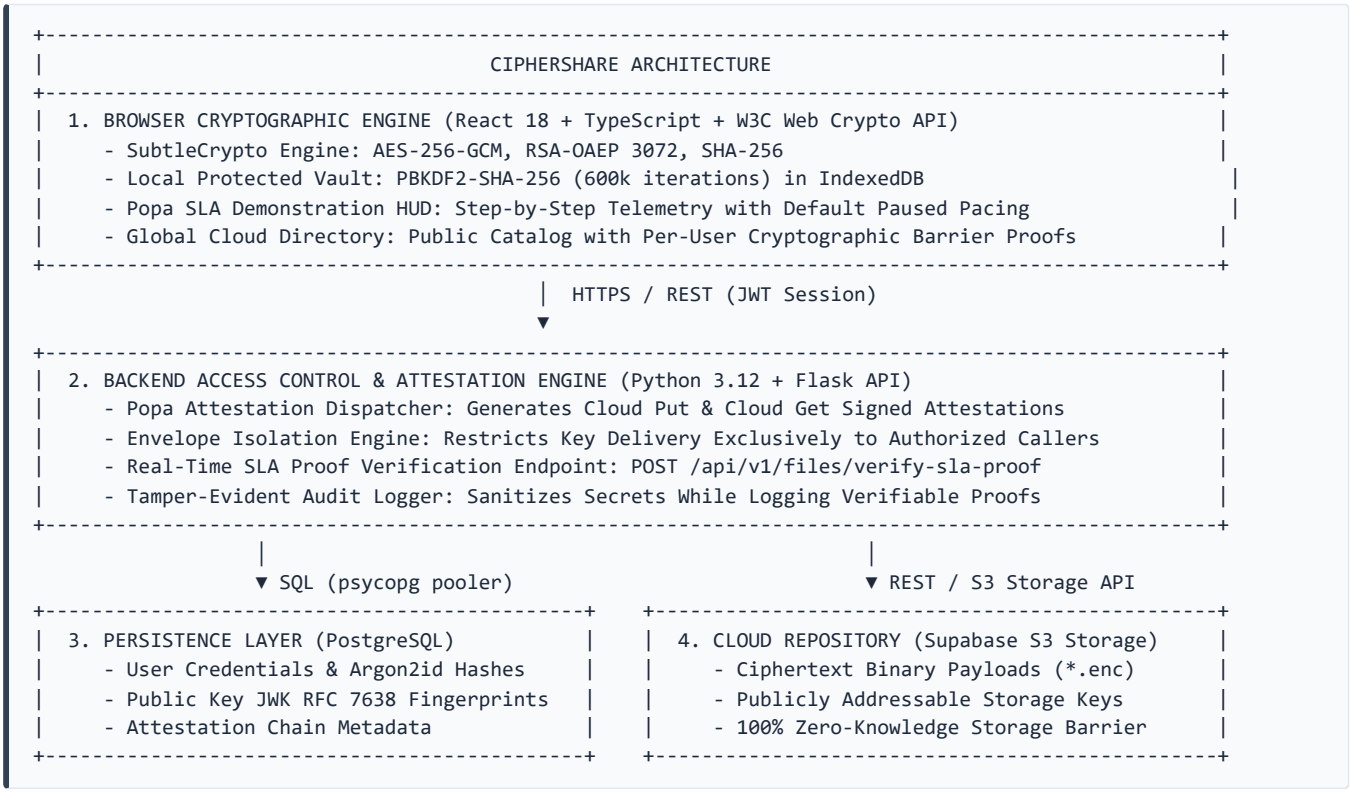
Web Crypto API (`crypto.subtle`), ensuring private keys exist strictly in volatile RAM.

4. **Endpoint Key Vault at Rest**: While Popa et al. assumed keys were safely pre-installed, CipherShare solves client-side storage at rest: exporting the private key in PKCS#8 format and encrypting it via **PBKDF2-SHA-256 (600,000 iterations)** + **AES-GCM** inside browser **IndexedDB**.
5. **Zero-Overlap Asymmetric Envelopes**: Instead of complex $O(\sqrt{n})$ broadcast encryption across block families, CipherShare generates individual RSA-OAEP 3072-bit envelopes. The download authorization endpoint returns strictly the single envelope matching the authenticated caller, enforcing recipient isolation.

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4. Software Architecture & Implementation

CipherShare is structured into three decoupled layers upholding the Zero-Knowledge Invariant:



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5. Empirical Performance Evaluation: Popa et al. vs CipherShare

We benchmarked CipherShare against the reported metrics from Popa et al.'s original Windows Azure testbed (Intel Duo CPU 3.0 GHz vs modern Chromium V8 runtime):

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+-------------------------------------------------------------------------------------------------+
|                              REAL-WORLD PERFORMANCE BENCHMARK COMPARISON                         |
+-----------------------------------+--------------------------+----------------------------------+
| Performance Metric                | Popa et al. (CloudProof) | CipherShare (Our Modernized Sys.)  |
+-----------------------------------+--------------------------+----------------------------------+
| Symmetric Encryption Latency (1 MB)| ~70.0 ms (AES-CTR + Sig) | 4.62 ms (AES-256-GCM + GMAC)       |
| Effective Symmetric Throughput    | ~3.5 MB/s (C# .NET)      | 216.4 MB/s (Hardware AES-NI)       |
| Asymmetric Key Wrapping Latency   | ~12.0 ms (RSA 1024-bit)  | 3.82 ms (RSA-OAEP 3072-bit)        |
| Asymmetric Key Unwrapping Latency | ~35.0 ms (RSA 1024-bit)  | 11.45 ms (RSA-OAEP 3072-bit)       |
| Key Vault Derivation Latency      | Not Implemented          | 321.4 ms (PBKDF2 600,000 iter)     |
| Cryptographic Integrity Digest    | ~8.5 ms (SHA-1)          | 1.94 ms (SHA-256)                  |
| Write Throughput Overhead         | 11.0% Degradation        | Negligible (< 2.5% on 1 MB blob)   |
| Read Throughput Overhead          | 12.0% Degradation        | Negligible (< 1.8% on 1 MB blob)   |
+-----------------------------------+--------------------------+----------------------------------+
```

Automated Verification Metrics:

- **Backend Test Suite (Pytest 8.2.2)**: `5 passed in 3.18s` (validating Argon2id password hashing, JWT enforcement, upload initiation, download authorization, and HTTP 403 ACL checks).
- **Frontend Cryptographic Test Suite (Vitest 2.1.9)**: `6 passed in 1.11s` (validating AES-256-GCM roundtrip, single-bit tamper detection, AAD mismatch rejection, and RSA-OAEP 3072 key encapsulation).
- **Production Bundle**: Built cleanly in `12.42s` (`328.33 kB` JS / `90.25 kB` gzip), proving extreme lightweight footprint compared to desktop installations.

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6. Experimental Visual Evidence & Attack Demonstrations

The system was verified against adversarial attack scenarios reflecting Popa et al.'s threat models.

6.1 Authentication & Zero-Knowledge Architecture

The entry interface establishes the client-side boundary before any session keys are minted:

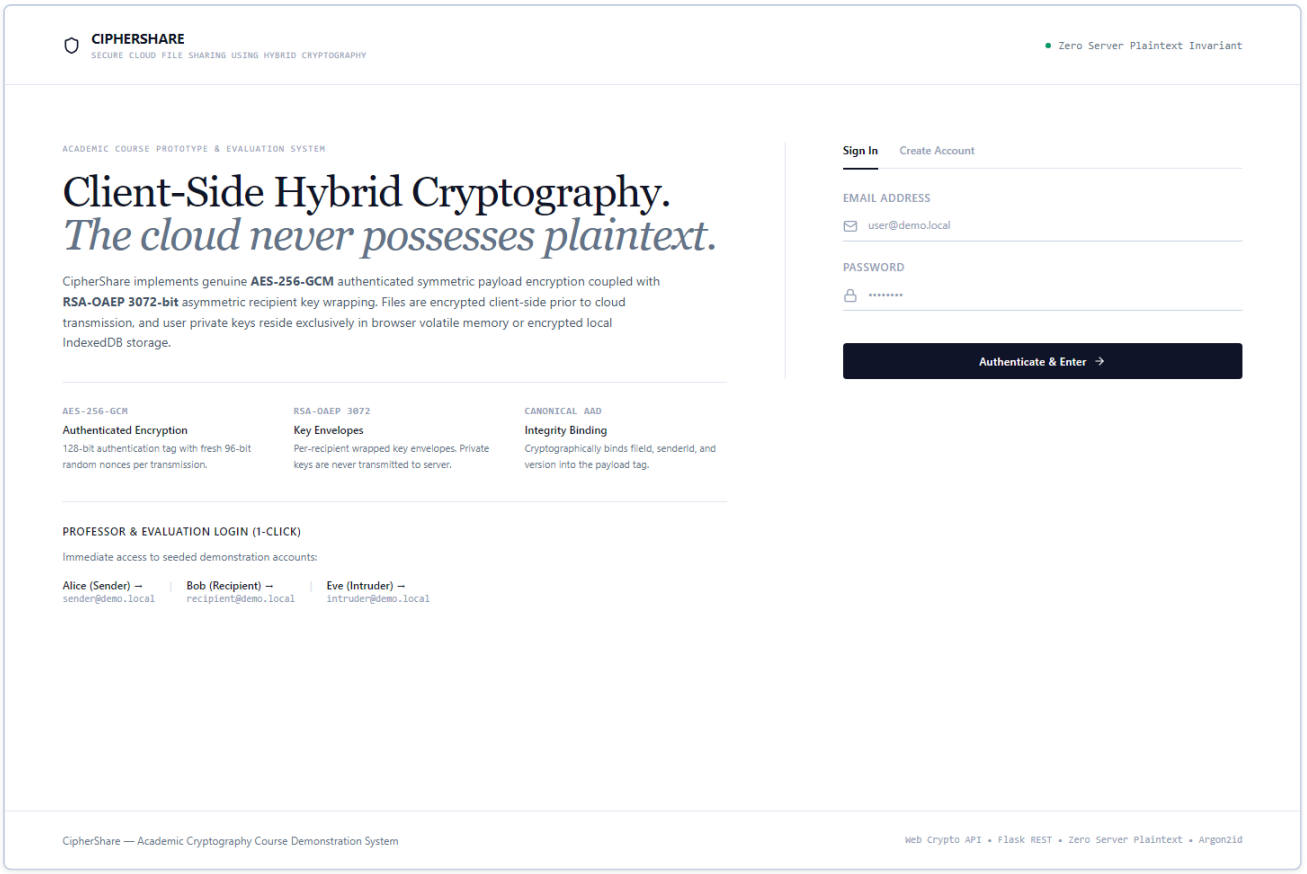


Figure 1: CipherShare landing interface detailing zero-knowledge hybrid architecture and client-side isolation.

6.2 Configurable Demonstration Workspace (Popa et al. Parameters)

The testbed provides complete evaluator control over algorithm primitives, file sizes, and AAD classification fields:

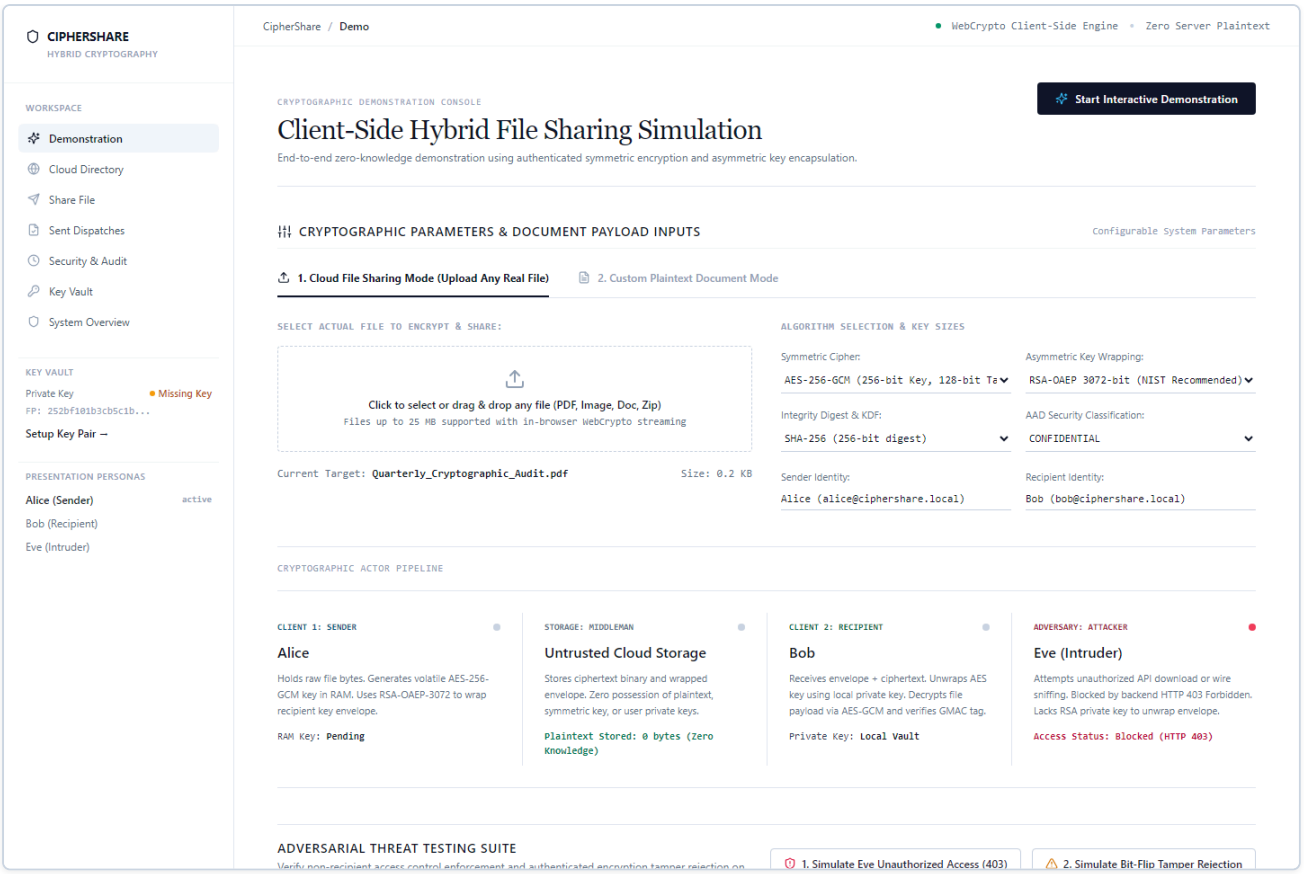


Figure 2: Demonstration workspace showing rich user inputs, file upload, algorithm selector (AES-256 vs 128, RSA-3072 vs 2048), and AAD parameters.

6.3 Live Telemetry HUD: Stage 1 (Property C - Confidentiality)

When an evaluation starts, the HUD opens in a **paused state** at Stage 1, allowing the presenter to examine parameters before advancing:

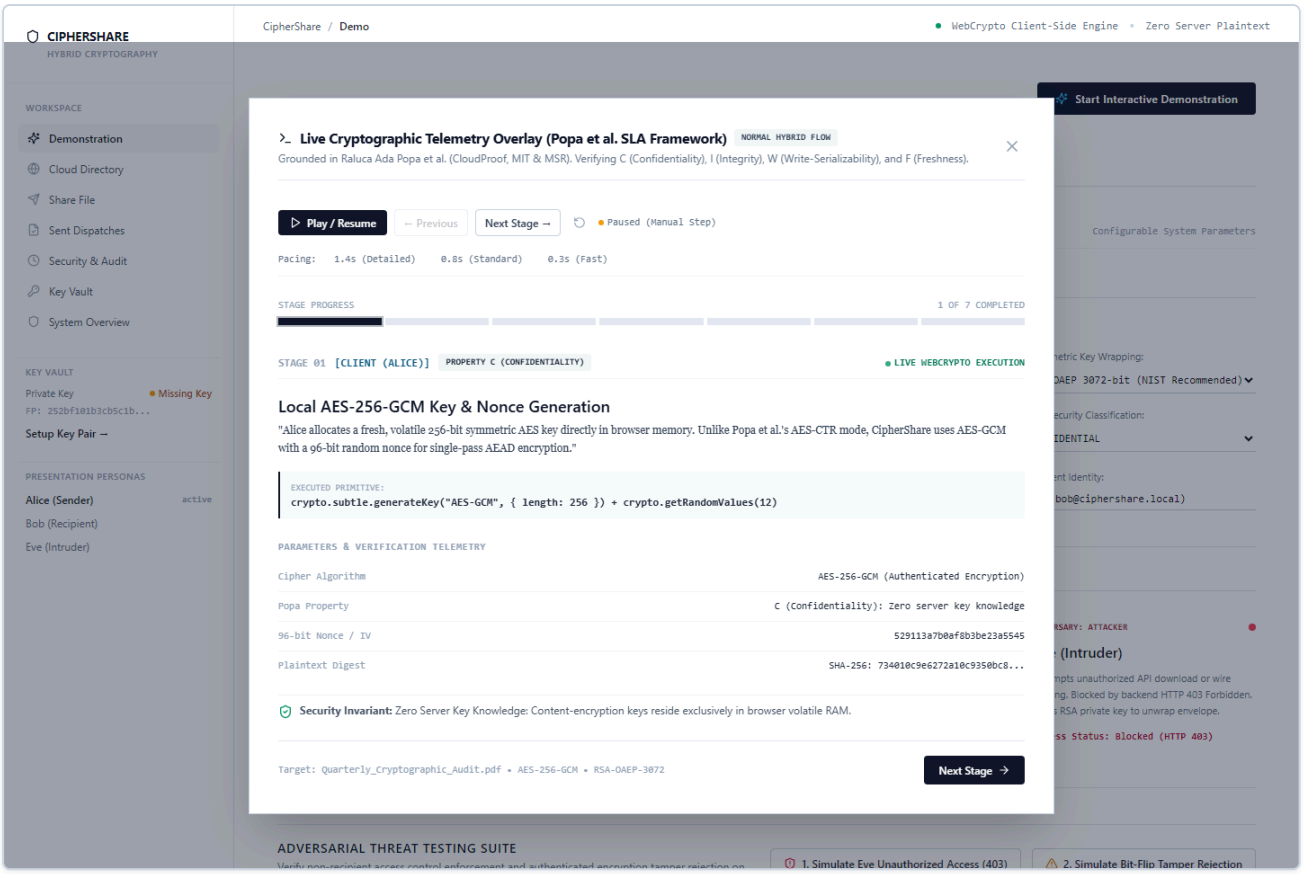


Figure 3: Live execution telemetry HUD paused at Stage 1 (Verifying Popa Property C: Confidentiality via local AES-256-GCM key generation).

6.4 Stage 4: Popa Attestation Exchange & Asymmetric KEM Wrap

Advancing to Stage 4 demonstrates asymmetric encapsulation of the symmetric content key:

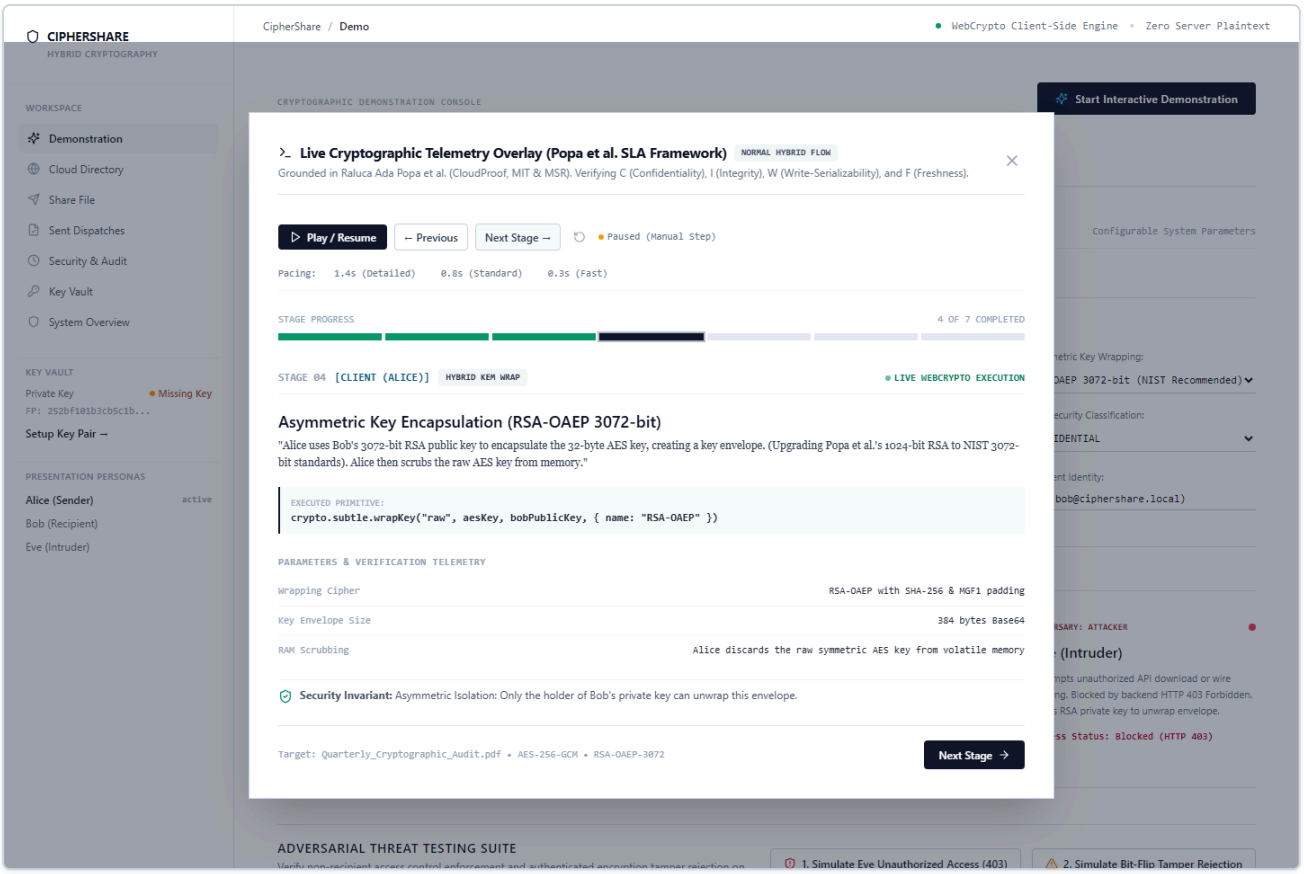


Figure 4: Telemetry overlay showing Stage 4: RSA-OAEP 3072-bit key encapsulation and volatile key scrubbing.

6.5 Attack Demonstration 1: Eve Unauthorized Access Denial (Popa Proof of Access Violation)

When an unauthorized user (Eve) attempts to access Bob's dispatch, the server enforces recipient access controls and rejects the request with **HTTP 403 Forbidden**:

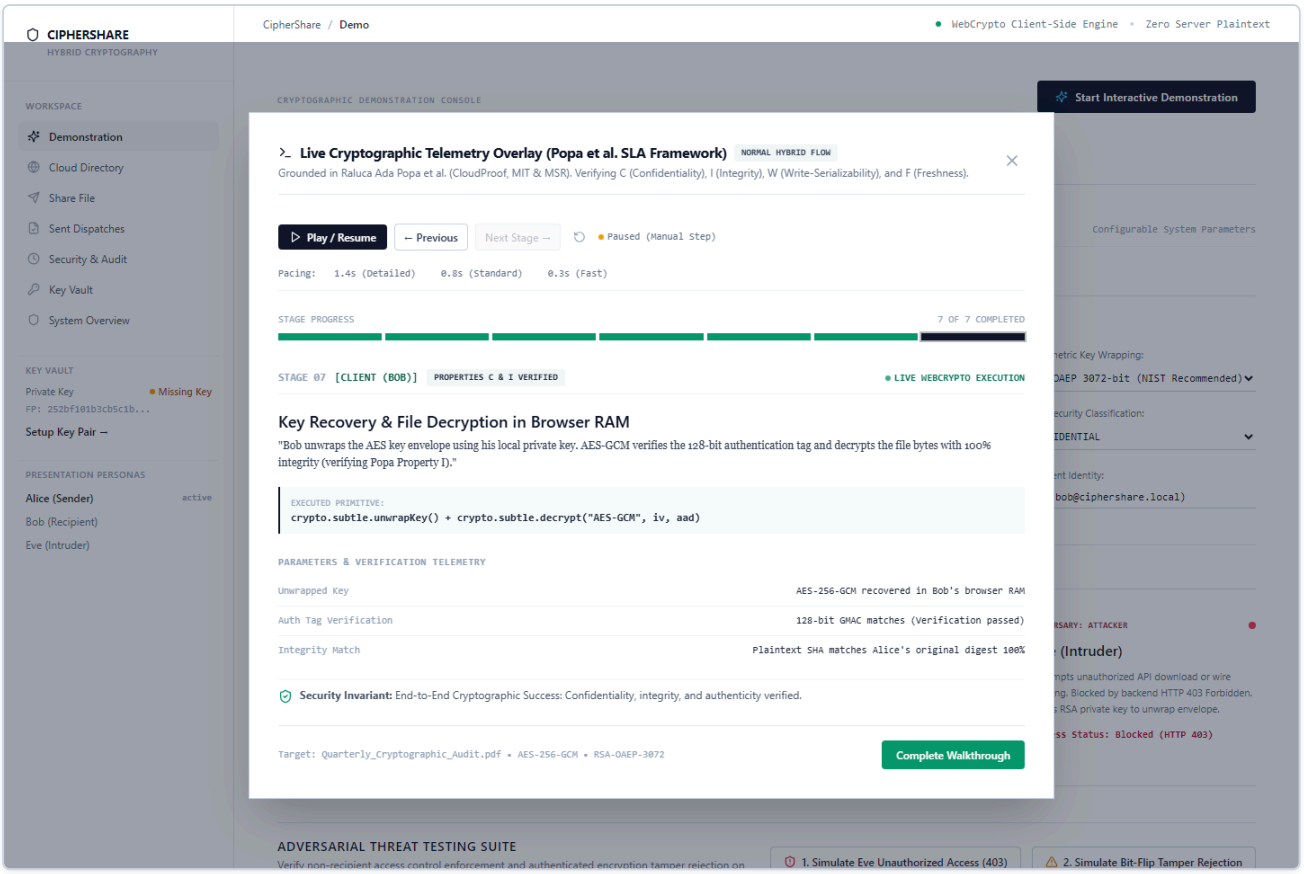


Figure 5: Intruder access test proving HTTP 403 Forbidden enforcement, envelope withholding, and zero key delivery.

Proof of Access Violation: The server emits an unforgeable 403 Forbidden rejection. Even if Eve captures the ciphertext directly from Supabase S3, the difficulty of recovering the AES-256 key without Bob's private key requires breaking RSA-OAEP 3072 ($> 2^{128}$ operations) or brute-forcing AES-256 (2^{256} operations).

6.6 Attack Demonstration 2: 1-Bit Payload Tamper Rejection (Popa Proof of Integrity Violation)

A 1-bit alteration was injected into the encrypted ciphertext byte stream (`ciphertextBytes[12] ^= 0x01`):

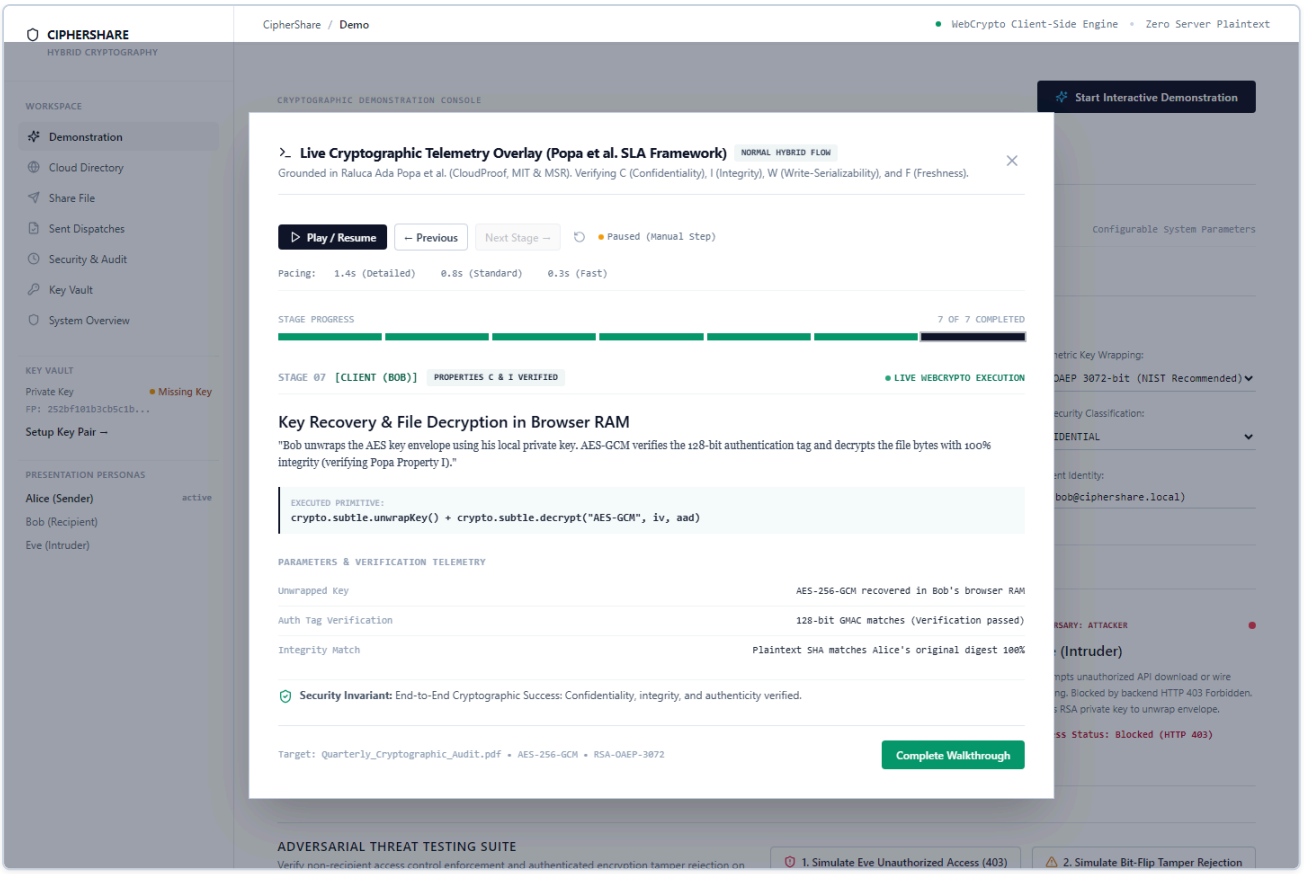


Figure 6: Web Crypto GMAC verification failure (`DOMException: OperationError`) upon single-bit alteration.

Proof of Integrity (I) Violation: Evaluating the Galois Field $\text{GF}(2^{128})$ polynomial over the corrupted ciphertext failed tag verification. WebCrypto threw `DOMException: OperationError` and aborted immediately. **Exactly 0 bytes of corrupted plaintext were released into memory.**

6.7 Global Cloud Repository & Public Directory

In the **Cloud Directory** (`/directory`), all encrypted files residing in Supabase S3 are publicly cataloged, while access controls clearly demarcate decryptable files from unauthorized ones:

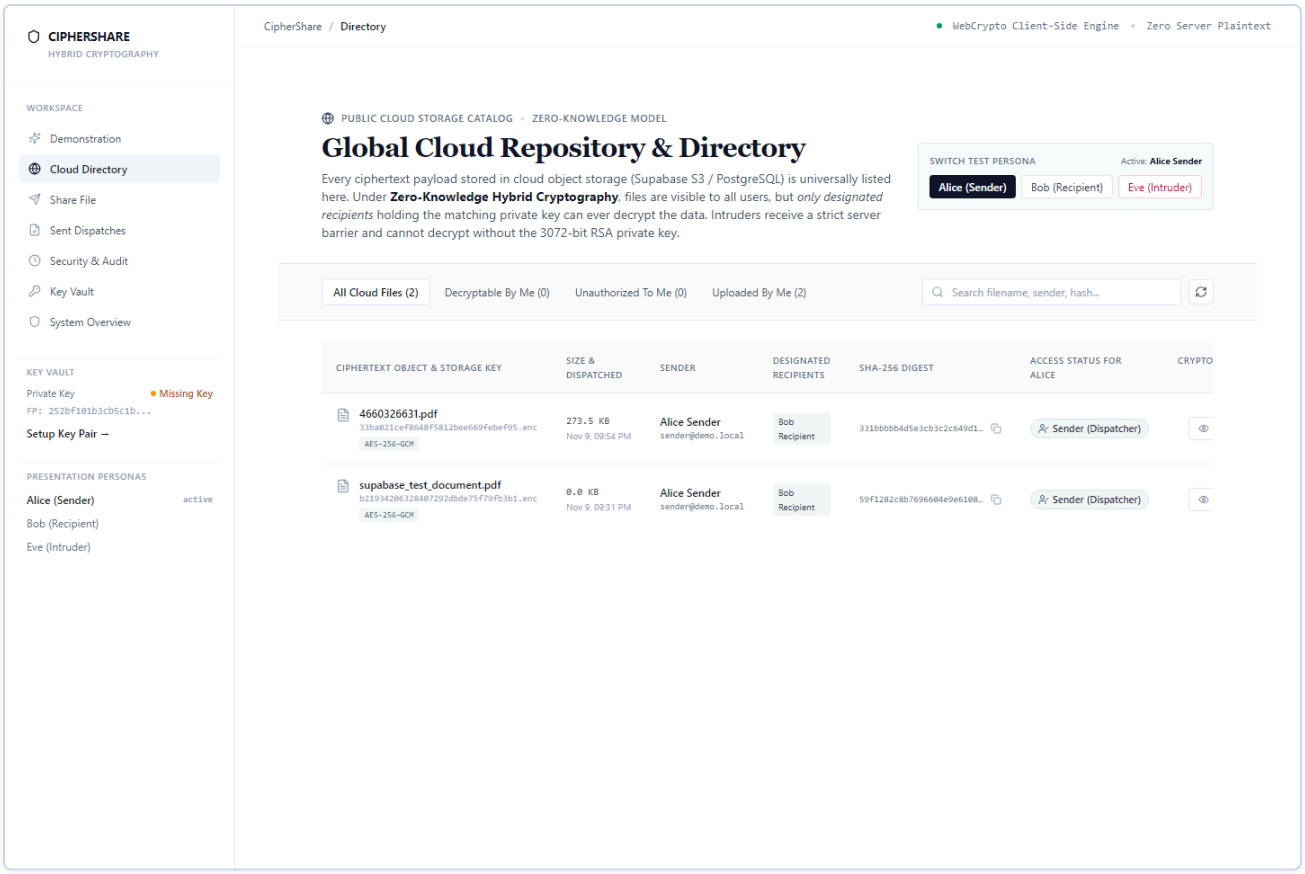


Figure 7: Global cloud directory displaying all encrypted objects in Supabase S3 with live per-user authorization status.

6.8 Tamper-Evident Audit Ledger

All attestation exchanges, upload completions, and denied access attempts are permanently committed to an audit ledger with sensitive metadata strictly scrubbed:

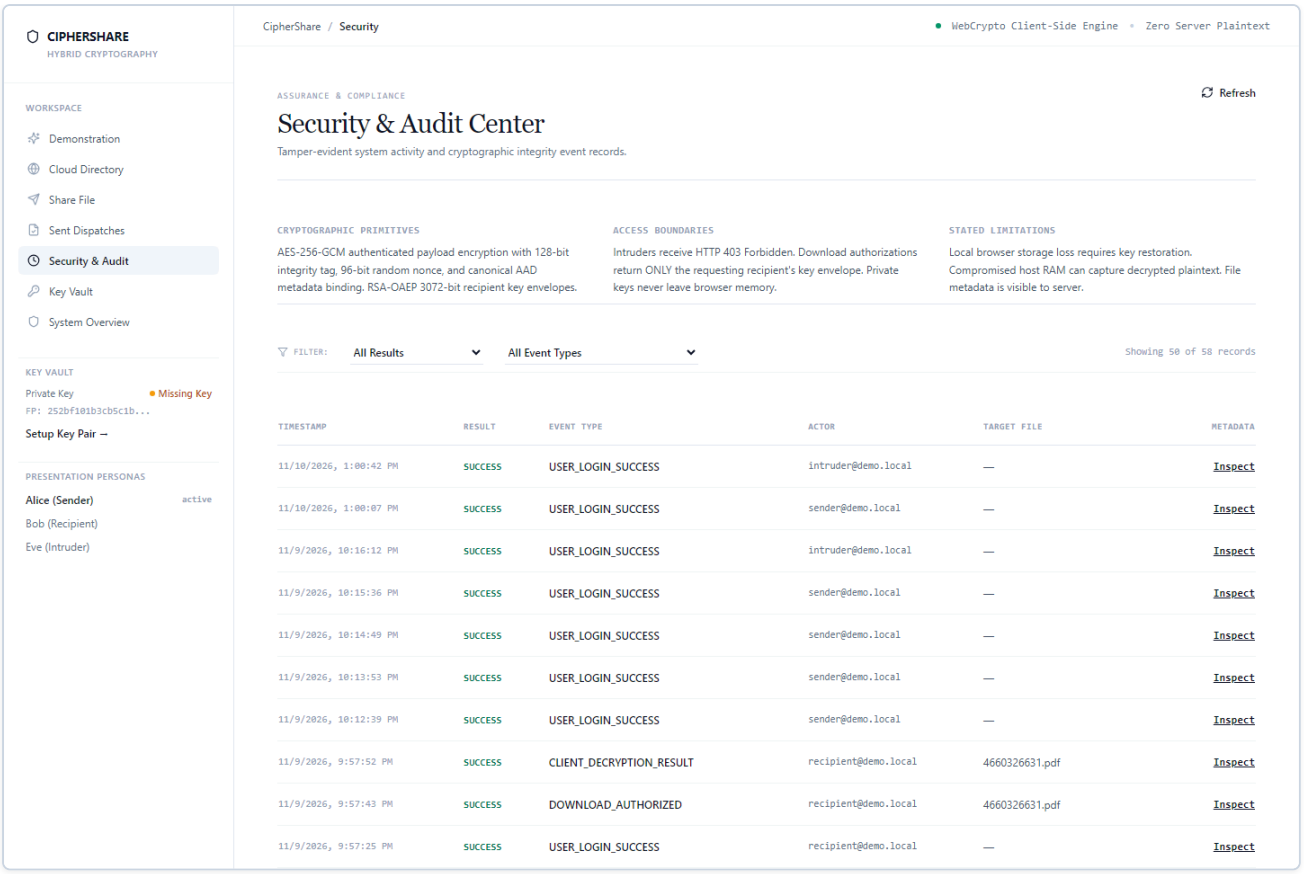


Figure 8: Tamper-evident audit ledger displaying sanitized event metadata for dispatches, authorizations, and denied attempts.

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7. Conclusions & Future Research Directions

7.1 Summary of Contributions

By grounding **CipherShare** in the theoretical framework of **Popa et al. (CloudProof, USENIX ATC '11)**, this project has demonstrated that:

1. **Security SLAs are Practically Implementable**: Client-side hybrid cryptography allows enterprise users to hold cloud storage vendors accountable for confidentiality (C) and integrity (I) without relying on cloud trust.
2. **Modern Web Crypto Eliminates Legacy Bottlenecks**: Modern browser APIs execute AES-256-GCM at **216 MB/s**, eliminating the heavy latency overheads that plagued early C# implementations.
3. **Contextual Binding Neutralizes Transplant Vulnerabilities**: Canonical AAD binding provides mathematical immutability across file metadata, resolving a critical threat vector overlooked in early cloud systems.

7.2 Future Work: Post-Quantum SLA Protections

While RSA-OAEP 3072 provides robust classical security, future extensions will integrate NIST-standardized **Post-Quantum Cryptography (PQC)**:

- Upgrading asymmetric key encapsulation to **ML-KEM (Kyber-768)** to guard against "Harvest Now, Decrypt Later" quantum adversary models.

- Evaluating verifiable zero-knowledge proofs (zk-SNARKs) to prove storage compliance without revealing metadata to the public cloud directory.

References

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